

Experiment – 2

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, NOT NULL

```
MySQL 8.0 Command Line Client
mysql> create database collegedb;
Query OK, 1 row affected (0.02 sec)

mysql> use collegedb;
Database changed
mysql> create table department(
  -> deptid int primary key,
  -> deptname varchar(50) not null);
Query OK, 0 rows affected (0.07 sec)

mysql> insert into department
  -> values
  -> (1,'cse'),
  -> (2,'ai&ds'),
  -> (3,'ece');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'department
values
(1,'cse'),
(2,'ai&ds'),
(3,'ece')' at line 1
mysql> insert into department
  -> values
  -> (1,'cse'),
  -> (2,'ai&ds'),
  -> (3,'ece');
Query OK, 3 rows affected (0.04 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> create table student(
  -> rollno int primary key,
  -> name varchar(50) not null,
  -> email varchar(50) unique,
  -> age int check(age >= 18),
  -> city varchar(30) default 'chennai',
  -> deptid int,
  -> foreign key
  -> foreign key(deptid) references department(deptid));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'key(deptid) references department(deptid))' at line 9
mysql> CREATE TABLE Student (
  -> RollNo INT PRIMARY KEY,
  -> Name VARCHAR(50) NOT NULL,
  -> Email VARCHAR(50) UNIQUE,
  -> Age INT CHECK (Age >= 18),
  -> City VARCHAR(30) DEFAULT 'Chennai',
  -> DeptID INT,
  -> FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
  -> );
Query OK, 0 rows affected (0.06 sec)
```

```
Select MySQL 8.0 Command Line Client
mysql> RollNo INT PRIMARY KEY
  -> Name VARCHAR(50) NOT NULL
  -> Email VARCHAR(50) UNIQUE
  -> Age INT CHECK (Age >= 18)
  -> City VARCHAR(30) DEFAULT 'Chennai'
  -> FOREIGN KEY (DeptID) REFERENCES Department(DeptID);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'RollNo INT PRIMARY KEY
Name VARCHAR(50) NOT NULL
Email VARCHAR(50) UNIQUE
Age IN' at line 1
mysql> INSERT INTO Student
  -> (RollNo, Name, Email, Age, DeptID)
  -> VALUES
  -> (101,'Rahul','rahu@gmail.com',20,1),
  -> (102,'Priya','priya@gmail.com',19,2),
  -> (103,'Arun','aru@gmail.com',21,3);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+-----+
| RollNo | Name | Email | Age | City | DeptID |
+-----+-----+-----+-----+-----+-----+
| 101 | Rahul | rahu@gmail.com | 20 | Chennai | 1 |
| 102 | Priya | priya@gmail.com | 19 | Chennai | 2 |
| 103 | Arun | aru@gmail.com | 21 | Chennai | 3 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Student
  -> VALUES
  -> (104,'Ramu','ramu@gmail.com',20,'Chennai',1);
ERROR 1062 (23000): Duplicate entry 'rahu@gmail.com' for key 'student.Email'
mysql> INSERT INTO Student
  -> VALUES
  -> (105,'Ravi','ravi@gmail.com',16,'Chennai',2);
ERROR 3819 (HY000): Check constraint 'student_chk_1' is violated.
mysql> INSERT INTO Student
  -> VALUES
  -> (106,'Anu','anu@gmail.com',20,'Chennai',10);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('collegedb`.`student`, CONSTRAINT `student_ibfk_1` FOREIGN KEY (`DeptID`) REFERENCES `department` (`deptid`))
mysql> DESCRIBE Student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RollNo | int | NO | PRI | NULL | |
| Name | varchar(50) | NO | | NULL | |
| Email | varchar(50) | YES | UNI | NULL | |
| Age | int | YES | | NULL | |
| City | varchar(30) | YES | | Chennai | |
| DeptID | int | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

